

Supplementary Materials

Anonymous CVPR submission

Paper ID 11838

1. Proof of Theorem 1

Proof. The existence of a minimizer θ_λ^* in Eq. 2 is easy. Take $\delta = g(\theta_\lambda^*)$. Then for any θ with $g(\theta) \leq \delta$, it holds that

$$f(\theta_\lambda^*) + \lambda g(\theta_\lambda^*) \leq f(\theta) + \lambda g(\theta) \leq f(\theta) + \lambda \delta$$

from which we deduce that $f(\theta_\lambda^*) \leq f(\theta)$.

For $0 < \lambda < \mu$, we have

$$f(\theta_\lambda^*) + \mu g(\theta_\lambda^*) \geq f(\theta_\mu^*) + \mu g(\theta_\mu^*)$$

$$f(\theta_\lambda^*) + \lambda g(\theta_\lambda^*) \leq f(\theta_\mu^*) + \lambda g(\theta_\mu^*)$$

from which we further deduce that

$$\mu (g(\theta_\mu^*) - g(\theta_\lambda^*)) \leq f(\theta_\lambda^*) - f(\theta_\mu^*) \leq \lambda (g(\theta_\mu^*) - g(\theta_\lambda^*)).$$

It is easy to see from the above inequalities that $g(\theta_\mu^*) \leq g(\theta_\lambda^*)$ and hence $f(\theta_\mu^*) \geq f(\theta_\lambda^*)$. $\square$

2. Additional Qualitative Results

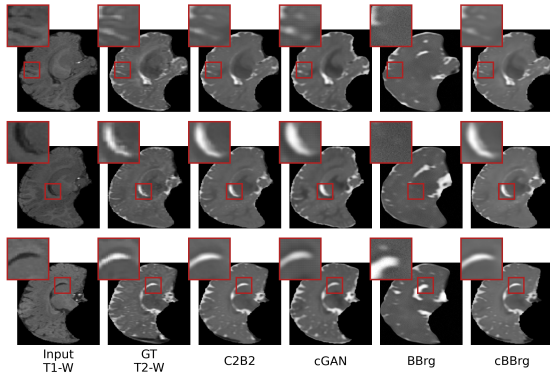


Figure 1. Additional qualitative results of iSEG 2017 dataset: T1-W to T2-W modality translation. Highlighted regions demonstrate local-level features for comparison across methods. The same applies to the remaining figures.

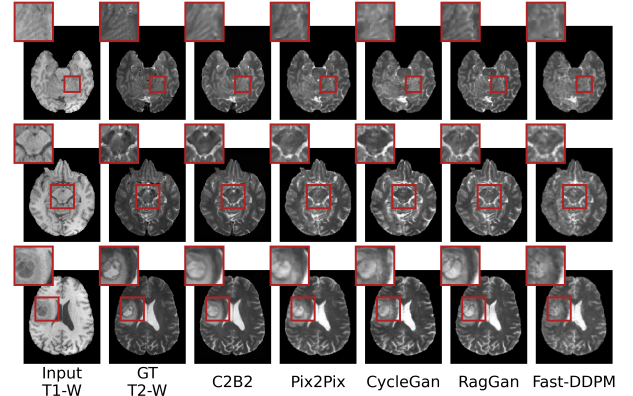


Figure 2. Additional comparison of C2B2 and SOTA Methods on the BraTS2018 T1-W to T2-W Image Translation Dataset. The images generated by Fast-DDPM method were sampled five times (only one sample is shown), resulting in a std. of 0.0071.

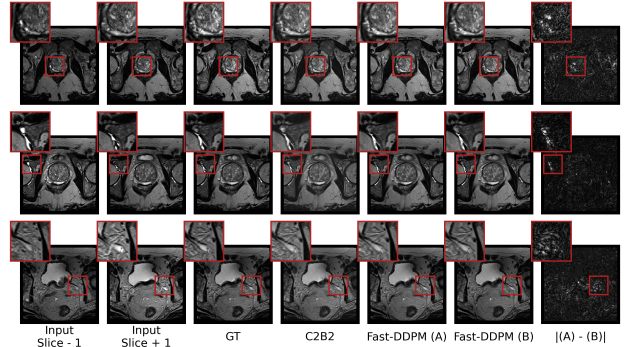


Figure 3. Additional comparison between C2B2 and SOTA methods on the Prostate MRI super-resolution dataset. The images generated by Fast-DDPM method were sampled five times, resulting in a std. of 0.0116. Two samples are shown with an absolute difference plot highlighting variations due to the non-deterministic approach.